Section 1

Wednesday, January 14, 2026 1:46 PM

- Outline of Topics

1) Moment generating function (MGF)

$M_X(t) = E(e^{tx})$

- uniquely defines a probability distribution
- moments:
  1. $E(X) = M_X'(0)$
  2. $E(X^2) = M_X''(0)$
  3. $E(X^3) = M_X'''(0)$
  ⋮
  k. $E(X^k) = M_X^{(k)}(0)$
- Taylor Expansion: $e^x = \sum_{k=0}^{+\infty} \frac{x^k}{k!}$
- $M_X(t) = E(e^{tx}) = E\left[\sum_{k=0}^{+\infty} \frac{t^k x^k}{k!}\right] = E\left[1 + tx + \frac{t^2x^2}{2} + \frac{t^3x^3}{6} + \cdots\right.$

$$\frac{\partial}{\partial t} E(e^{tx}) = E\left[x + tx^2 + \frac{t^2x^3}{2} + \cdots\right]\Big|_{t=0} = E(x)$$

$$\frac{\partial^2}{\partial^2 t} E(e^{tx}) = E\left[x^2 + tx^3 + \cdots\right]\Big|_{t=0} = E(x^2)$$

- continuous R.V.

$E(X) = \int x \cdot f(x)\,dx$

$E(X^2) = \int x^2 \cdot f(x)\,dx$

⋮

$E(X^k) = \int x^k \cdot f(x)\,dx$

- discrete R.V.

$E(X) = \sum x \cdot p(x)$

2) Expectation and Variance of Random Variables

$E(X) = \mu = \int_x x \cdot f(x)\,dx$

$Var(X) = E(X^2) - E(X)^2$

3) Evaluating Integrals by Recognizing "Kernels"

First, integration by parts:

$\int u\,dv = vu - \int v\,du$

- pdf: $f(x) = \underbrace{(\text{normalizing constant})}_{\text{constant w.r.t. } x}\underbrace{(\text{kernel})}_{\text{varies w.r.t. } x}$

ex: $X \sim N(\mu, \sigma^2)$

$f(x) = (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$, $-\infty \le x \le +\infty$

← $N(\mu, \sigma^2)$ Kernel

$$f(x) = (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \quad , -\infty \le x \le +\infty$$

$\leftarrow N(\mu, \sigma^2)$ Kernel

ex: $X \sim \text{Gamma}(\alpha, \beta)$

(shape, scale representation)

$$f(x) = \frac{\beta^{-\alpha}}{\Gamma\alpha} x^{\alpha-1} \exp\left\{-\frac{x}{\beta}\right\} \quad , x \ge 0$$

$\leftarrow G(\alpha, \beta)$ Kernel

- Fact: Integration over the pdf of a RV's entire support evaluates to 1

$X \sim \text{Gamma}(\alpha, \beta)$

$$\int_0^{+\infty} \frac{\beta^{-\alpha}}{\Gamma\alpha} x^{\alpha-1} \exp\left\{-\frac{x}{\beta}\right\} dx = 1 \quad \star$$

$$E(X) = \int_0^{+\infty} x \cdot \frac{\beta^{-\alpha}}{\Gamma\alpha} x^{\alpha-1} \exp\left\{-\frac{x}{\beta}\right\} dx$$

$$= \frac{\beta^{-\alpha}}{\Gamma\alpha} \int_0^{+\infty} \underbrace{x^{\alpha} \exp\left\{-\frac{x}{\beta}\right\}}_{G(\alpha+1, \beta) \text{ Kernel}} dx$$

$$= \frac{\beta^{-\alpha}}{\Gamma\alpha} \cdot \frac{\Gamma(\alpha+1)}{\beta^{-(\alpha+1)}} \cdot \int_0^{+\infty} \frac{\beta^{-(\alpha+1)}}{\Gamma(\alpha+1)} x^{\alpha} \exp\left\{-\frac{x}{\beta}\right\} dx \quad \nearrow 1$$

$$= \frac{\cancel{\beta^{-\alpha}} \cdot \alpha \cancel{\Gamma\alpha}}{\cancel{\Gamma\alpha}\, \cancel{\beta^{-\alpha}}\, \beta^{-1}} = \boxed{\alpha\beta}$$